* eccentricity

Alan bazlı filtreleme

Morphological image Processing

↳ binary

↳ grayscale

Z^2 ↔ iki boyutlu tamsayı uzayı

$Z^2_{x,y}$

↑ x, noktası

$A = \{(a_1, b_1), (a_2, b_2), \dots\}$

$= \{(x_1, y_1), (x_2, y_2), \dots\}$

$a = (a_1, a_2)$

$a \in A$

$a \notin A$ ← elemanı değil

$A \subseteq B$

A, B'nin alt kümesi

$C = A \cup D$ ← birleşimi

$D = A \cap B$ ← ortak elemanlar

$A \cap B \neq \emptyset$ ← A ve B ayrık

$A^c = \{w \mid w \notin A\}$

w öyleki A'da değil

$A - B = \{w \mid w \in A, w \in B\} = A \cap B^c$

$\hat{B} = \{w \mid w = -b, \text{ for } b \in B\}$

↑ reflection

$(A)_z = \{w \mid w = a + z, \text{ for } a \in A\}$

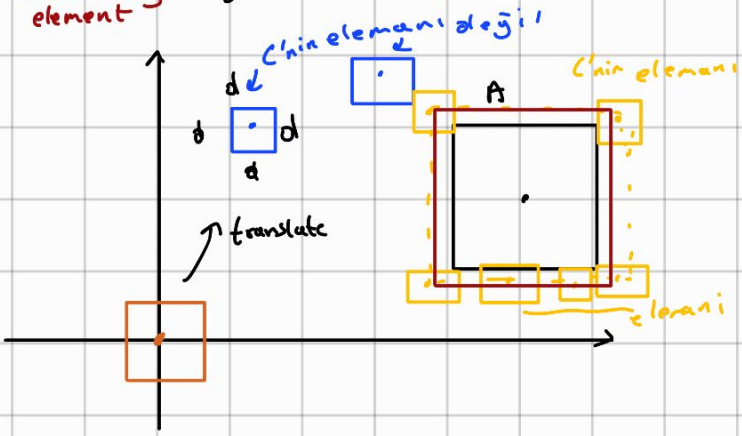
↑ A kümesini z noktasına translation

Dilation operator

A and B sets in Z^2

$C = A \oplus B$ → structuring element $= \{z \mid (\hat{B})_z \cap A \neq \emptyset\}$

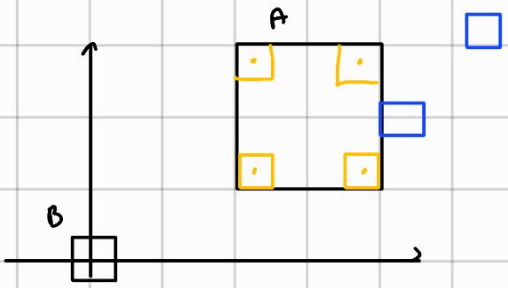
A dilayted by B



A'yı d kadar büyütün dilayet etin

Erosion:

$C = A \ominus B = \{z \mid (B)_z \subseteq A\}$



$(A \ominus B)^c = A^c \oplus \hat{B}$

$A \ominus B = (A^c \oplus \hat{B})^c$

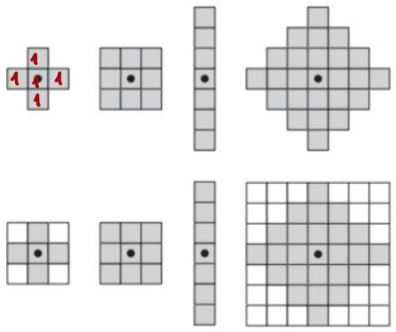
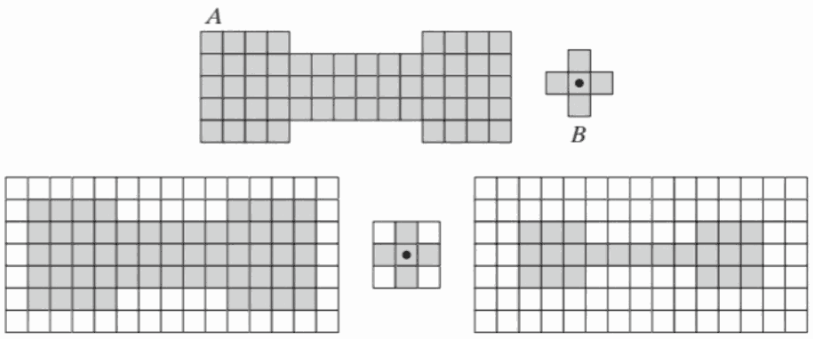


FIGURE 9.2 First row: Examples of structuring elements. Second row: Structuring elements converted to rectangular arrays. The dots denote the centers of the SEs.

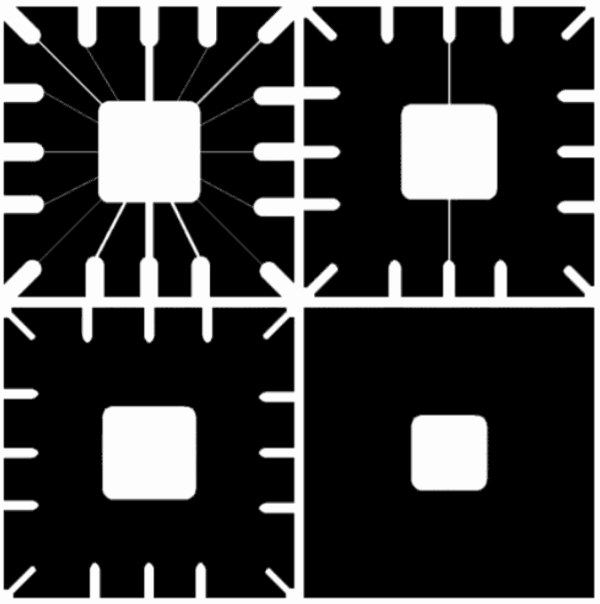


a b
c d e

FIGURE 9.3 (a) A set (each shaded square is a member of the set). (b) A structuring element. (c) The set padded with background elements to form a rectangular array and provide a background border. (d) Structuring element as a rectangular array. (e) Set processed by the structuring element.

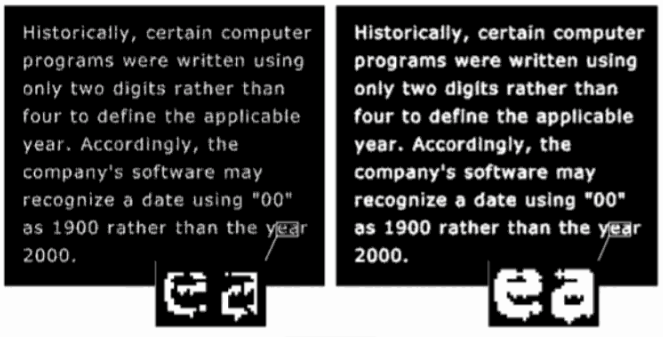
a b
c d

FIGURE 9.5 Using erosion to remove image components. (a) A 486×486 binary image of a wire-bond mask. (b)–(d) Image eroded using square structuring elements of sizes 11×11, 15×15, and 45×45, respectively. The elements of the SEs were all 1s.



a c
b

FIGURE 9.7
(a) Sample text of poor resolution with broken characters (see magnified view). (b) Structuring element. (c) Dilation of (a) by (b). Broken segments were joined.



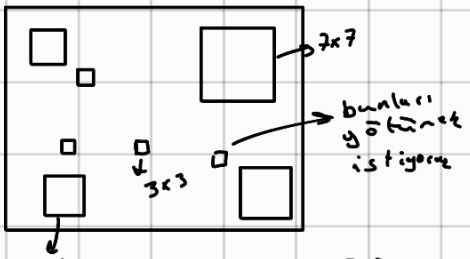
0	1	0
1	1	1
0	1	0

Opening and Closing

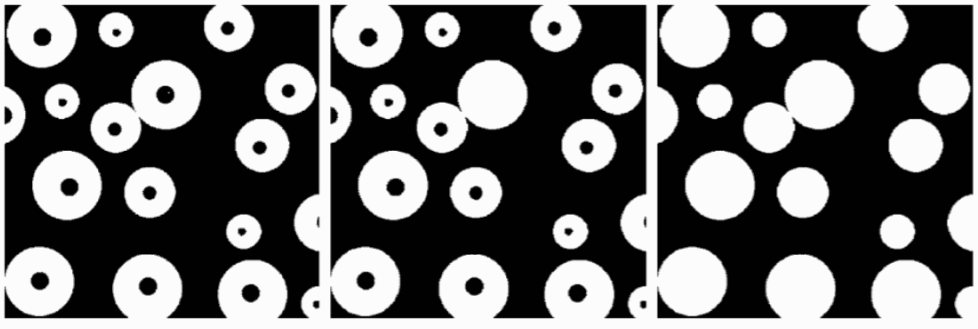
opening $A \circ B = (A \ominus B) \oplus B$ (önce küçültüp sonra büyütüyor)

closing $A \bullet B = (A \oplus B) \ominus B$

$(A \bullet B)^c = (A^c \circ B^c)$ (aslında bu görüntünün negatifi)



opening yaparsak 3x3'ü yok ederiz



a b c

FIGURE 9.16 (a) Binary image (the white dot inside one of the regions is the starting point for the hole-filling algorithm). (b) Result of filling that region. (c) Result of filling all holes.

Closing delikleri kapatıyor



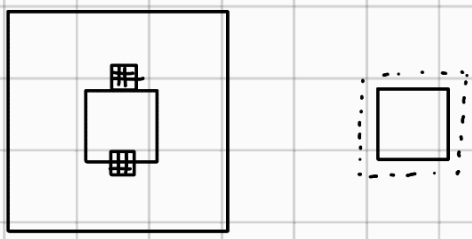
a b
d c
e f

FIGURE 9.11
(a) Noisy image.
(b) Structuring element.
(c) Eroded image.
(d) Opening of A.
(e) Dilation of the opening.
(f) Closing of the opening.
(Original image courtesy of the National Institute of Standards and Technology.)

&

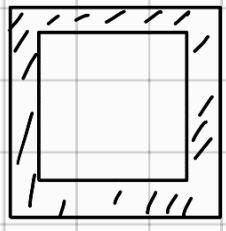
$A - (A \ominus B)$ → inner edge

$(A \oplus B) - A$ → outer edge



$(A \oplus B) - (A \ominus B)$

↳ inner, outer edge



Hit-OR-Miss Transform

Hit-And-Miss aslında

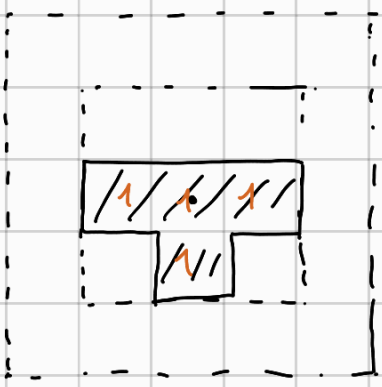
$$A \circledast B = \underbrace{(A \ominus X)}_{\text{hit}} \cap \underbrace{[A^c \ominus (W - X)]}_{\text{Miss}}$$

$B = (X, W)$

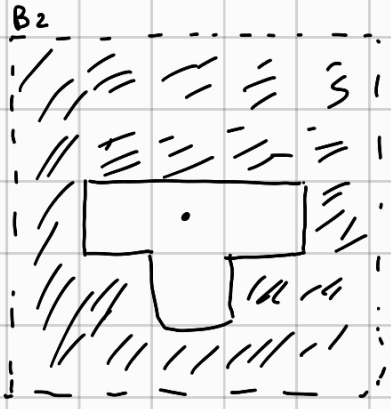
$A \circledast B = (A \ominus B_1) \cap [A^c \ominus B_2]$

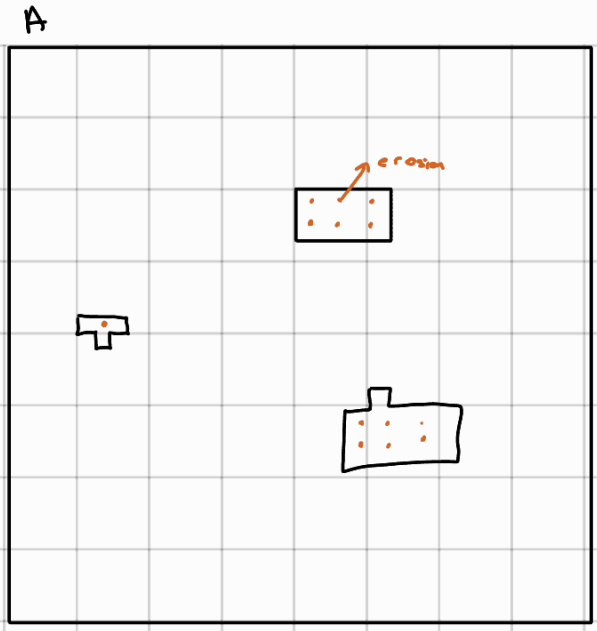
$B = (B_1, B_2)$

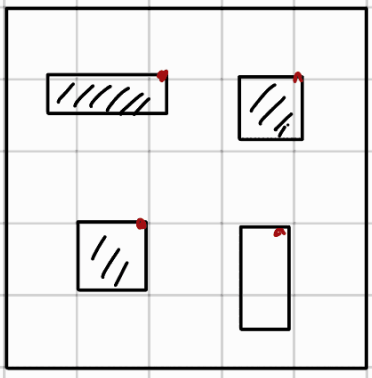
B_1

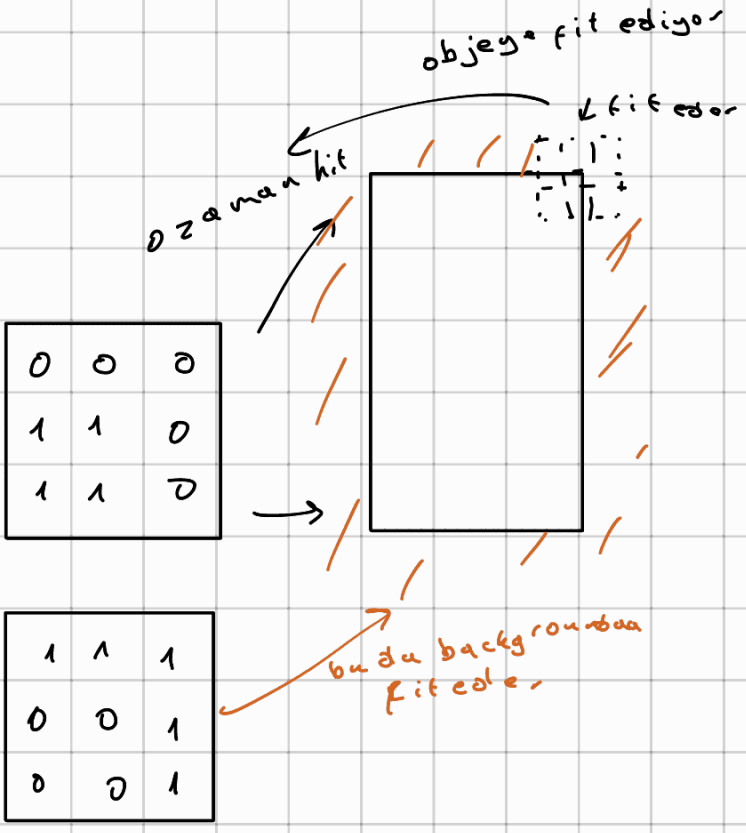


B_2









Hole filling

$$X_k = (X_{k-1} \oplus B) \cap I^c \qquad k = 1, 2, 3, \ldots$$

if $X_k = X_{k-1} \Rightarrow$ stop

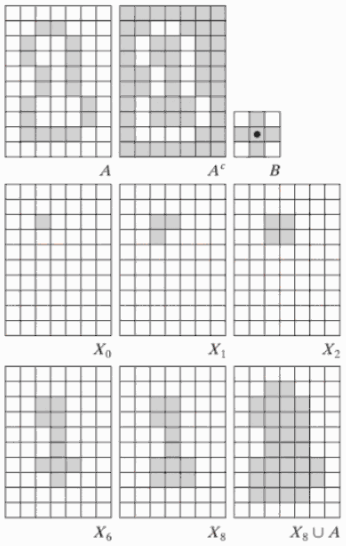
delikten bir nokta ile başlamak gerekiyor.

$$X_1 = (X_0 \oplus B) \cap A^c$$

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a b c
d e f
g h i

FIGURE 9.15 Hole filling. (a) Set A (shown shaded). (b) Complement of A. (c) Structuring element B. (d) Initial point inside the boundary. (e)–(h) Various steps of Eq. (9.5-2). (i) Final result [union of (a) and (h)].



$$A \oplus (B \oplus C) = (A \oplus B) \oplus C$$

$$A \ominus (\underbrace{B \oplus C}_{D}) = (A \ominus B) \ominus C$$

$$A \oplus B = B \oplus A$$

$$((A \circ B) \circ B) \circ B = A \circ B$$

$$(A \circ B) \circ B = A \circ B$$